CHECKING UNIVERSAL CLOSEDNESS IN THE NOETHERIAN LIMIT CASE

THE STACKS PROJECT AUTHORS

Editorial note. No omissions: the (4)=>(2) specialization construction is fully worked; the other directions are exact citations of proved lemmas.

Allowed standard prerequisites (editorial).

- Valued point lift along affines. Limits, lift lemmas of the chapter

Source reference. Original equation 0206: sources/stacks-chow/tags.txt, source lines 2623–2623. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 05JX: sources/stacks-chow/tags.txt, source lines 7181–7181. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01KF: sources/stacks-chow/tags.txt, source lines 2048–2048. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01K9: sources/stacks-chow/tags.txt, source lines 2042–2042. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 054F: sources/stacks-chow/tags.txt, source lines 6638–6638. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00PH: sources/stacks-chow/tags.txt, source lines 931–931. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : X \to S$ be a finite type morphism of schemes. Assume S is locally Noetherian. Then the following are equivalent*

(1) *f is universally closed,*
(2) *for every n the morphism $\mathbf{A}^n \times X \to \mathbf{A}^n \times S$ is closed,*
(3) *for any diagram (0206) there exists some dotted arrow,*
(4) *for all diagrams (0206) with A a discrete valuation ring there exists some dotted arrow.*

Proof. The equivalence of (1) and (2) is a special case of Lemma 05JX. The equivalence of (1) and (3) is a special case of Schemes, Proposition 01KF. Trivially (3) implies (4). Thus all we have to do is prove that (4) implies (2). We will prove that $\mathbf{A}^n \times X \to \mathbf{A}^n \times S$ is closed by the criterion of Schemes, Lemma 01K9. Pick n and a nontrivial specialization $z \rightsquigarrow z'$ of points in $\mathbf{A}^n \times S$ and a point $y \in \mathbf{A}^n \times X$ lying over z. Note that $\kappa(y)$ is a finitely generated field extension of $\kappa(z)$ as $\mathbf{A}^n \times X \to \mathbf{A}^n \times S$ is of finite type. Hence by Properties, Lemma 054F or Algebra, Lemma 00PH implies that there exists a discrete valuation ring $A \subset \kappa(y)$

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with fraction field $\kappa(z)$ dominating the image of $\mathcal{O}_{\mathbf{A}^n \times S, z'}$ in $\kappa(z)$. This gives a commutative diagram

$$\begin{array}{ccccc} \mathrm{Spec}(\kappa(y)) & \longrightarrow & \mathbf{A}^n \times X & \longrightarrow & X \\ \downarrow & & \downarrow & & \downarrow \\ \mathrm{Spec}(A) & \longrightarrow & \mathbf{A}^n \times S & \longrightarrow & S \end{array}$$

Now property (4) implies that there exists a morphism $\mathrm{Spec}(A) \to X$ which fits into this diagram. Since we already have the morphism $\mathrm{Spec}(A) \to \mathbf{A}^n$ from the left lower horizontal arrow we also get a morphism $\mathrm{Spec}(A) \to \mathbf{A}^n \times X$ fitting into the left square. Thus the image $y' \in \mathbf{A}^n \times X$ of the closed point is a specialization of y lying over z'. This proves that specializations lift along $\mathbf{A}^n \times X \to \mathbf{A}^n \times S$ and we win. $\square$